MCO-1D COMMISSIONING MANUAL — v13 ADDENDUM

Per-wing AoA sensing, stall detection, and closed-loop control

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This addendum covers changes since Revision 12. It adds one section on the per-wing AoA sensing hardware and software, updates the commissioning test sequences to include sensor calibration, and documents the implications for all affected systems.

SECTION 50 — PER-WING AoA SENSING SYSTEM

50.1 Why Individual Wing AoA Matters

The previous stall protection system used global airspeed and commanded AoA to estimate when any wing might be stalling. This was adequate for conventional flight but inadequate for the novel hypotheses that deliberately command very different AoA on different wings simultaneously. H-25 commands port wings near stall while starboard wings are at low AoA. H-26 commands large AoA differentials between sides. H-24 sets up a fore-to-aft AoA gradient. In all these cases, the system previously had no way to know whether the commanded AoA was actually achieved or whether any individual wing had entered stall independently.

50.2 The MCP3208 ADC

One MCP3208 8-channel 12-bit SPI ADC reads the feedback potentiometer output of all eight pitch servos. Each pitch servo has a feedback pin that outputs a voltage proportional to its actual shaft position. Reading this voltage through the ADC gives the actual angle of attack of each wing independently at 50 Hz. The resolution is approximately 0.044 degrees per ADC step — finer than the servo can position itself.

ADC chip	MCP3208, 8-channel, 12-bit, SPI interface
Connection	SPI Bus 1, CE1. Pi GPIO: CLK=11, MOSI=10, MISO=9, CS=7
Sample rate	All 8 channels read in ~107µs per 50Hz tick
Resolution	12-bit / 3.3V reference → 0.8mV per step → ~0.044°
Calibration	Per-servo during TC-01: three-point voltage-to-angle mapping
Fallback	If ADC not available: system uses Tiers 1, 2, 4 only (per-pair)

50.3 Installing the MCP3208

Enable SPI Bus 1 by adding to /boot/config.txt:

```
dtoverlay=spi1-1cs
```

Connect the chip as shown in BOM v13 addendum wiring diagram. Add a 1kΩ series resistor on each feedback signal line as protection. After connection, verify the ADC is readable with:

```
python3 -c "import mco_wave_library as w; print(w.read_servo_feedback_adc())"
```

If the MCP3208 is not fitted, this returns [None]*8 and the system falls back to per-pair stall detection gracefully.

50.4 Servo Feedback Calibration (TC-01 addition)

For each pitch servo, command three known positions and record the ADC voltage at each. Then call:

```
calibrate_servo_feedback(wing_idx=0, v_at_neutral=1.65, v_at_max=2.45, v_at_min=0.55,
deg_max=90, deg_min=-90)
```

Repeat for all 8 wings (wing_idx 0-7 = W1p, W1s, W2p, W2s, W3p, W3s, W4p, W4s). The calibration is stored in the Wave Library and survives across flights via the knowledge export system.

● CAUTION: Do not skip servo feedback calibration before flying hypotheses H-24, H-25, H-26, or H-27. These configurations rely on the closed-loop AoA controller which requires accurate calibration to command the correct aerodynamic forces.

SECTION 51 — CAPABILITIES AUGMENTED BY PER-WING AoA FEEDBACK

51.1 AoA Closed-Loop Control

Previously AoA was open-loop: command 18°, assume 18° is delivered. The pitch servo holds the wing against aerodynamic loads that vary continuously through the flapping stroke. At high frequency or high AoA, the servo may lag behind the commanded position. The closed-loop controller reads actual AoA every tick and applies a PID correction to the commanded angle to close the gap. The effect: the wing actually flies at the commanded AoA rather than approximately near it.

Kp = 0.30	***CALIBRATE TC-02*** — proportional gain
Ki = 0.05	***CALIBRATE TC-02*** — integral gain (persistent error)
Kd = 0.02	***CALIBRATE TC-02*** — derivative gain (damps oscillation)
Max corr.	5° per tick — prevents large sudden corrections

✓ NOTE: Conservative gains are set initially. After TC-02 the actual servo response time is known and gains can be tuned for tighter tracking.

51.2 Per-Wing Lift Estimation

With known actual AoA, airspeed, and frequency, the system estimates lift per wing using the S1223 polar. This enables: checking actual trim balance between port and starboard, validating that H-24 actually achieves the intended thrust-lift split, detecting lift asymmetry before it becomes a control problem, and providing the optimizer with actual lift data rather than lift estimated from commanded AoA.

```
lift_balance_report (shows per-wing lift and port/stbd balance)
```

51.3 Optimizer Trial Validity

Each discovery trial is now checked for AoA tracking quality. If the mean AoA error across all wings exceeds 3 degrees during a trial, the trial is marked invalid and discarded from the optimizer scoring. This removes the largest source of noise in the exploration process: trials that scored poorly not because the configuration was bad, but because the servos did not achieve the commanded position.

51.4 Servo Health Monitoring

The rolling RMS of AoA tracking error per wing is tracked continuously. A healthy servo holds within 0.5 degrees of commanded. A degrading servo shows increasing tracking error before it fails hard. This gives the same kind of trend-based early warning for servos that motor current already provides for motors.

```
servo_health_report (shows tracking RMS per wing, health status)
```

Below 2.0°	Healthy — tracking within tolerance
2.0° to 5.0°	CAUTION — tracking error rising, inspect servo
Above 5.0°	WARNING — servo may be failing, land when practicable

51.5 H-27 — Servo Bandwidth Mapping

A new hypothesis enabled entirely by the MCP3208 feedback. At 3.0 Hz flapping each stroke lasts 333 milliseconds. If the pitch servo takes 150 milliseconds to move from one AoA to another, it arrives at the correct position with only 183 milliseconds of stable AoA during the power stroke. Without feedback this was completely unknown. H-27 maps per-wing servo tracking speed across four frequencies and three airspeeds, giving the optimizer a model of actual AoA authority available at each operating point.

```
disc_mode_engage_focused("biomimetic") # H-27 tests in biomimetic block
```

51.6 Stall Detection Resolution

With servo feedback, stall detection upgrades from per-pair to per-wing. When W2 port stalls during an H-25 per-stroke AoA roll test, the system detects W2 port specifically, reduces only W2 port AoA, and leaves W2 starboard at its commanded position. Without feedback this distinction was impossible — both wings of the pair would have been corrected together, cancelling the deliberate asymmetry that makes H-25 work.

51.7 Active Damping Accuracy

The active damping system uses differential amplitude to stop residual roll. The stopping force depends on actual AoA of the damping wings — if aerodynamic loads have pushed those wings to higher-than-commanded AoA, the damping force is larger than predicted and may overshoot. With feedback: the active damping controller computes actual force from actual AoA and adjusts amplitude accordingly, preventing overshoot and producing crisper hesitation.

SECTION 52 — TC-01 ADDITIONS FOR AoA FEEDBACK CALIBRATION

Add these test steps to TC-01 after the servo range verification:

TC-01-12	MCP3208 ADC verification	python3 -c "import mco_wave_library as w; print(w.read_servo_feedback_adc())" Must print 8 float values (not None). If all None: check SPI wiring.
TC-01-13	Servo feedback calibration — W1p	Command W1p to neutral (0°). Read ADC voltage. Command to +45°. Read voltage. Command to -45°. Read voltage. Call calibrate_servo_feedback(0, v_neutral, v_plus, v_minus, 45, -45). Verify round-trip: command 20°, read back — must be within 1°.
TC-01-14	Repeat for all 8 pitch servos	Calibrate W1s (1), W2p (2), W2s (3), W3p (4), W3s (5), W4p (6), W4s (7). Export knowledge after calibration: knowledge_export_usb.
TC-01-15	Closed-loop tracking verification	With wing stationary, command step AoA changes. Verify closed-loop controller converges to commanded value within 5 ticks (0.1 seconds). If oscillating: reduce Kp. If slow: increase Kp.
TC-01-16	Lift balance check (static)	Command all wings to same AoA (15°). Run lift_balance_report. Port and starboard should be within 5%. Larger asymmetry indicates a calibration error on one servo.